

Breaking Ciphers & Fingerprinting AI

A statistical mindset from classical cryptanalysis to modern AI

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github.com/JamieGrunewald/cipher-ai-analysis

"A strange game. The only winning move is not to play."

— WarGames, 1983

1 9 8 3 .

A teenager dials into NORAD.
A supercomputer named Joshua answers.
It wants to play a game.

LOGON: FALKEN

PASSWORD: _

IDENTIFICATION NOT RECOGNIZED

ACCESS CODE: CPE1704TKS

GREETINGS, PROFESSOR FALKEN.

SHALL WE PLAY A GAME?

CPE1704TKS — Joshua's launch authorisation code. We're going to crack it.

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Bug Bounty — Bugcrowd / HackerOne

This Talk

Course

CPEG 472 Applied Cryptography
University of Delaware

Prof

Prof. Andrew Novocin

Project

Graduate cryptanalysis assignment
Shotgun hill-climbing from scratch

Repo

github.com/JamieGrunewald/cipher-ai-analysis

> REMEMBER WHEN THE INTERNET WAS FUN? _

> curiosity > credentials

> we broke things to understand them

> knowledge was shared, not gated

> structure always told a story

A G E N D A

Measure structure → Guide optimization → Generalize

01 Why Weak Ciphers Fail

Language structure leaks. Always.

02 Statistical Reconnaissance

Measure the structure

03 Hill-Climbing as Optimization

Follow the gradient

04 Shotgun Hill-Climbing

Escape local optima

05 The Bridge

Same mindset, new system

06 Something to Think About

Where else does structure leak?



SECTION 01

Why Weak Ciphers Fail

Language structure survives transformation

- *And anything that leaves structure behind... can be measured.*
- *Encryption hides symbols, but language has measurable statistical structure that leaks through any cipher that doesn't destroy it.*

Letter Frequencies

E=12.7%, T=9.1%, A=8.2% — substitution preserves these entirely

Index of Coincidence

Measures letter repeat probability — 0.065 for English, 0.038 for random

N-gram Patterns

TH, HE, IN — predictable pairings survive most transformations

Word Structure

ZIT ZIT → THE THE — positional repetition is a direct fingerprint

Grammatical Redundancy

Human language is ~50% redundant — structurally impossible to fully mask



SECTION 02

Statistical Reconnaissance

Know your target before you attack

0.065

English IOC

0.038

Random IOC

26!

Keyspace

χ^2

Fit metric

Step 1: classify the cipher before you attack

Wrong classifier = wrong tool. IOC tells you the cipher family in milliseconds.

IOC near 0.065 → monoalphabetic (Caesar / Substitution)
IOC near 0.038 → polyalphabetic (Vigenère)

→ Classify. Then choose your attack. Order matters.

English letter frequency (top 6)

E 12.7%

T 9.1%

A 8.2%

O 7.5%

I 7.0%

N 6.7%

Chi-Square (χ^2) — scores candidate plaintext fitness. Lower = closer to natural English.

SECTION 03

Hill-Climbing as Optimization

Cryptanalysis is a search problem

$26! \approx 4 \times 10^{26}$ possible keys — brute force is impossible

1

Random Key

Start with any key — position doesn't matter

2

Decrypt

Apply key, produce candidate plaintext

3

Score

χ^2 + n-gram log probability fitness

4

Mutate

Swap two letters in the key

5

Keep if Better

Accept improvements, discard regressions

6

Repeat

Converge toward correct key

Pseudocode

```
best_key = random_key()
best_score = score(decrypt(ct, best_key))

for _ in range(MAX_ITER):
    new_key = mutate(best_key)
    s = chi2(decrypt(ct, new_key))
    + ngram(decrypt(ct, new_key))
    if s > best_score:
        best_key = new_key
        best_score = s

return best_key
```

Why this works: structure leaks a gradient.

We're following that gradient—not searching randomly.

Every χ^2 improvement means we moved toward the real key. The signal exists because language is redundant.

Local optima trap this — hence shotgun restarts →

SECTION 04

Shotgun Hill-Climbing

Escaping local optima with parallel restarts

One hill-climber gets trapped. Many in parallel find the global peak.

Goal: understand why
— not just that it works

Ciphers targeted

Caesar / ROT-n

26 keys — χ^2 scoring cracks it trivially

Vigenère

Kasiski + IOC → key length → per-column Caesar

Substitution

Shotgun hill-climb with n-gram fitness

Hill Cipher

Known-plaintext + matrix algebra mod 26

Col. Transposition

Permutation search — work in progress

Demo output — substitution breaker

Ciphertext:

ZIT JXOEA WKGVF YGB PXDHL ZIT SQMN RGU

Attempt 001 score=-423.8 → JIBBERISH

Attempt 047 score=-201.3 → THE QVICK...

Attempt 112 score=-148.1 → THE QUICK...

Attempt 184 score=-98.6 ✓

Plaintext:

THE QUICK BROWN FOX JUMPS THE LAZY DOG

Key recovered. Time: 847ms.

Joshua encrypted his access attempts with his own launch code. Let's recover it.

1

IOC column sweep → key length = 6

k=6 peaks at IOC=0.059 / all other lengths < 0.050

2

Split into 6 columns → 6 independent Caesar ciphers

col[0] shift=2 → C	col[3] shift=19 → T
col[1] shift=15 → P	col[4] shift=10 → K
col[2] shift=4 → E	col[5] shift=18 → S

3

Word-list check confirms CPETKS

CPETKS decrypts → GREETINGS PROFESSOR FALKEN	✓
EPETKS decrypts → EREETILGS PRODESSOR DALKEN	✗

The ciphertext

key: CPE1704TKS (digits stripped → CPETKS)

GREETINGS PROFESSOR FALKEN
A STRANGE GAME THE ONLY
WINNING MOVE IS NOT TO PLAY
SHALL WE PLAY A GAME

→ IGIXDAPVW IBGHTWLYJ HPPDOF
C HXKKFIT KTWV VWI HXDA
LMGXAPV QHFW KH RHD LQ EPTI
ZQL EUYMV P RBMW IPQX

Recovered: CPETKS

Add back the digits →

CPE1704TKS

Joshua's launch code. We just found it.

// one more thing

The same mindset applies to AI text.

AI isn't encrypted. But it is produced by probabilistic optimization —
and probabilistic optimization always leaves measurable structure.

Goal: explore how text can be locally optimized — not attribute authorship. The question is whether structure leaks. Not who wrote it.

CIPHER ATTACK

Measure: letter freq, IOC

Probe: mutate key, score χ^2

Signal: language structure leaks



AI TEXT PROBE

Measure: perplexity, entropy

Probe: perturb tokens, score shift

Signal: optimization structure leaks

// something to think about

Every system that optimizes language leaves a statistical fingerprint.

So, what does that mean for the systems we haven't looked at yet?

Compilers:

Optimised machine code has statistical regularities. Does your compiler choice leak into your binary — and can an adversary read it?

Obfuscators:

Obfuscators try to destroy structure. How completely can they? What residue always survives, and what does that residue tell you?

LLM Agents:

When an AI rewrites its own output to evade a detector, is it running hill-climbing against a fitness function — and who controls that function?

Joshua said: *"The only winning move is not to play."* Was he wrong about everything?